SCHEME AND ANSWER KEY

MGU-BVOC (HONOURS) Regular EXAMINATION April 2026

Second Semester

SDC-MG2SDCREM102-MATHEMATICS FOR RENEWABLE ENERGY -2

Part A

Answer any 17 questions

Each carries 1 marks

1. Inverter ..... 1 mark [U] /[CO1]

2.



.....1 mark [U]/[CO1]

3. 1 and 0. .....1 mark [U]/[CO1]

4. $(x-2)^2 = 4a(y-5)$..... 1 mark [A]/[CO2]

5. (c+h , k) and (-c+h , k)..... 1 mark [A]/[CO2]

6. (0,0)..... 1 mark [A]/[CO2]

7. 2c.....1mark[A]/[CO2]

8. $(x+4)^2/a^2 - (y+8)^2/b^2 = 1$.....1 mark [A]/[CO2]

9. $2b^2/a = 2$.....1 mark [A]/[CO2]

10. Minimize the cost of final blend....1 mark [C]/[CO3]

11. Objective function has to be maximized subject to the given constraints....1 mark [C]/[CO3]

12. Minimum cost.....1 mark [C]/[CO3]

13. A cycle C_n, $n \geq 3$, consists of n vertices $v_1, v_2, \ldots, v_n$ and edges $\{v_1, v_2\}, \{v_2, v_3\}, \ldots, \{v_{n-1}, v_n\}$, and $\{v_n, v_1\}$. .....1 mark [U]/[CO4]

14. 

.....1mark[U]/[CO4]

15. For graph..... 0.5 mark

For it's subgraph...0.5 mark[U]/[CO4]

16. A graph with no loops and multiple edges.....1 mark [U]/[CO4]

17. n+1 .....1 mark [U]/[CO4]

Part B

Answer any 3 questions

Each carries 5 marks

18.

x	y	xy	x+y
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	1

.....3 marks

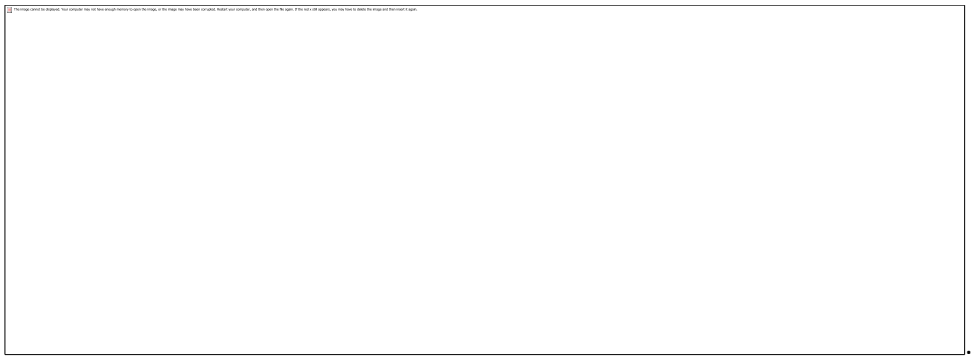
xy=x+y when x=y=0 and x= y= 1.....2 marks [U]/[CO1]

19. 2a=10. So a=5.....1mark

c=4.....1 mark

Eccentricity e=c/a.....1mark

e=4/5=0.8. .....2 marks [A]/[CO2]

20.  .....3 marks

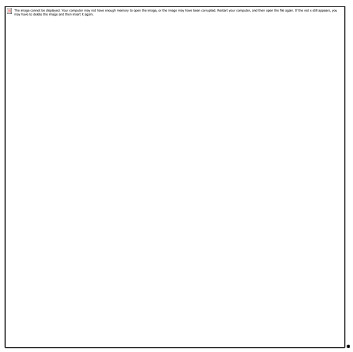
Corner points	Z=-x+2y
(3,2)	-3+2=1
(4,1)	-4+2=-2
(6,0)	-6+0=-6

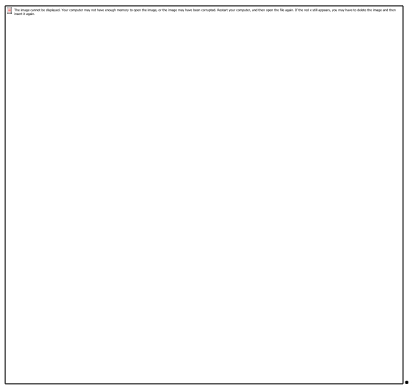
..... 2 marks [C]/[CO3]

21. Graph .....2 marks

Complement of the graph.....3 marks [U]/[CO4]

22

 .....3 marks

 .....2 marks [U]/[CO4]

Part. C

Answer any 2 questions

Each carries 10 marks

23.

x	y	z	xy	xz	xy+xz
0	0	0	0	0	0
0	0	1	0	0	0
0	1	0	0	0	0
1	0	0	0	0	0
1	0	1	0	1	1
1	1	0	1	0	1
0	1	1	0	0	0
1	1	1	1	1	1

.....10 marks [U]/[CO1]

24. $x^2 = 4ay$. .....1 mark

25=8a. 3mark

a=25/8. .....1 mark

x^2=4y25/8

=25y/2

$2x^2$= 25y. .....5 marks [A]/[CO2]

25. a). For full capacity

1.5x+3y =42

3x+y =24

Where x is number of rackets

Y is number of bats.

.....1 mark

Solving these equations

Number of rackets x=4

Number of bats y=12

.....4 marks

b) profit=20x+10y

=80+120

=200

.....5 marks [C]/[CO3]

26. Definition and example of incidence Matrix

Let G = (V, E) be an undirected graph. Suppose that $v_1, v_2, \dots, v_n$ are the vertices and $e_1, e_2, \dots, e_m$ are the edges of G. Then the incidence matrix with respect to this ordering of V and E is the n× m matrix M = [mij], wherewhere

mij ={1 when edge ej is incident with vi

0 other wise

.....5 marks

Definition and example of adjacency Matrix

Suppose that G = (V, E) is a simple graph where |V| = n. Suppose that the vertices of G are listed arbitrarily as $v_1, v_2, \dots, v_n$. The adjacency matrix A (or AG) of G, with respect to this listing of the vertices, is the n x n zero–one matrix with 1 as its (i, j)th entry when vi and vj are adjacent, and 0 as its (i, j)th entry when they are not adjacent. . .....5 marks [U]/[CO4]